

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING
CODE: MLA0408

COURSE

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:
20

Total Marks:

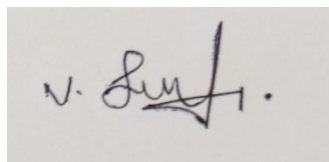
1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I Vemula saiteja (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

Date
6/8/26

Name : Vemula Sai Teja
Reg NO : 192425083

C02 - AT3

1) Machine Learning (ML)

Machine learning is a subset of Artificial intelligence where algorithms learn patterns from data to make predictions or decisions without being explicitly programmed.

Characteristics

- * uses statistical algorithms
- * Required manual feature engineering
- * works well with structured data
- * faster to train than deep learning
- * suitable for smaller datasets.

Deep learning (DL)

Deep learning is a subset of machine learning that uses Artificial Neural many hidden layers to automatically learn complex features from data.

Characteristic

- * learns feature automatically
- * performs well on unstructured data
- * Needs large dataset
- * Require GPUs/TPUs
- * can solve complex tasks like image recognition and language understanding

Comparison Table

Feature	Machine learning	Deep learning
Architecture	Simple algorithms	Multi-layer neural networks
Data requirement	Small to medium dataset	Very large dataset
Feature extraction	Manual	Automatic
Training time	Faster	Slower
Hardware	CPU sufficient	GPU/TPU preferred
Images & Audio	Limited	Excellent.

Advantages of ML

- * Faster training
- * Less computational cost
- * Easier interpretation

Advantages of DL

- * Manual feature extraction
- * Higher accuracy on complex tasks.

2.

Representation Learning

Representation learning is the process where a neural network automatically discovers useful features from raw data instead of relying on manually engineered features.

Types

1. supervised learning

Uses labeled data

* CNN

* Vision transformer

2. unsupervised learning

Uses unlabeled data

* Autoencoders

* Restricted Boltzmann machine

Example

Raw Image

↓
edges

↓
shapes

↓
objects

↓
Classification

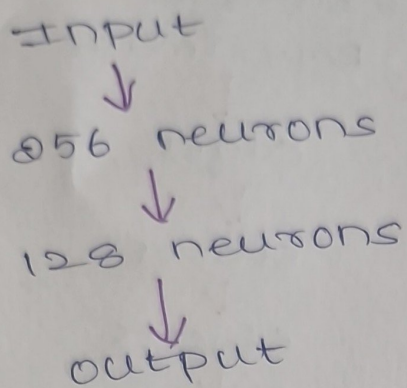
Importance.

1. Eliminates manual feature engineering
2. Improve accuracy
3. learns hierarchical feature
4. Generalizes better
5. Variational Autoencoders.

width of Neural Network

width means the number of neurons in each hidden layer

Example:-

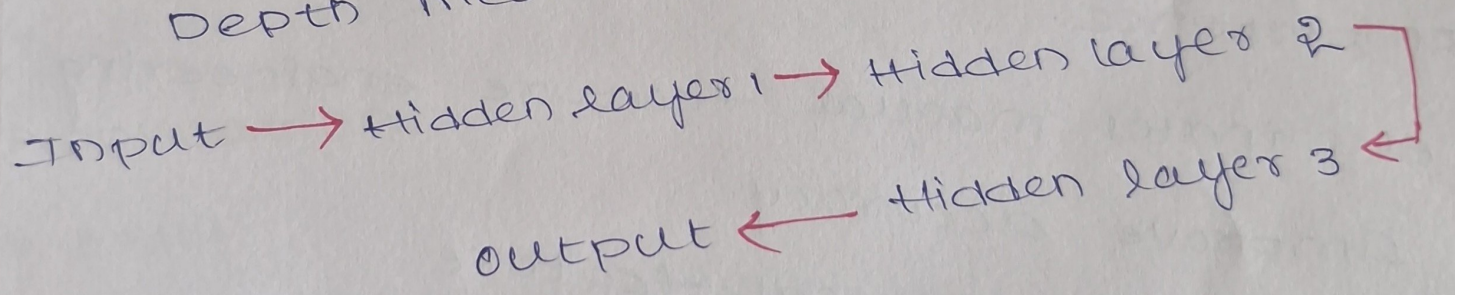


Advantages

1. learns more features
2. Better representation
3. Faster convergence some time.

Depth of Neural Network

Depth means the no. of hidden layers



Advantages

- * learns hierarchical representations
- * Better for image recognition
- * High accuracy
- * can solve complex problems.

3.

1. ReLU

Formula

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Graph

Negative $\rightarrow 0$

Positive $\rightarrow$ straight line

Advantages

- * Very fast
- * easy computation
- * prevents vanishing gradient
- * sparse activation

Disadvantages

- * Dying ReLU problem
- * zero gradient for negative inputs

Applications

- * CNN
- * image classification
- * object detection

2. Leaky ReLU

Formula

$$f(x) = \begin{cases} ax, & x < 0 \\ x, & x \geq 0 \end{cases}$$

where $\alpha = 0.01$

Advantages

- ⇒ solves dying ReLU
- ⇒ gradient exists for negative values
- ⇒ faster convergence

Disadvantages

- ⇒ α chosen manually
- ⇒ slightly higher computation

Applications

1. GANs
2. Deep CNNs
3. ELU (Exponential Linear Unit)

Formula

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha (e^x - 1), & x \leq 0 \end{cases}$$

Advantages

- * Negative output push mean activations closer to zero
- * Faster convergence
- * Reduces vanishing gradients compared with sigmoid / tanh

Disadvantages

- * More computationally expensive because of the exponential function
- * Slower than ReLU

Comparison table

Feature	ReLU	Leaky ReLU	ELU
Negative output	0	Small negative	Smooth negative
Dying Neuron	yes	No	No
Speed	Fastest	Fast	Moderate
Computation	Very low	low	Higher
Accuracy	Good	Better	Often best
Gradient for $x < 0$	0	Small	Exponential

4. Unsupervised learning

Unsupervised learning trains model using unlabeled data. The model identifies hidden pattern, structure, or relationships without knowing the correct outputs.

Goals

- * Clustering
- * Dimensionality reduction

- Feature extraction
- Anomaly detection

Advantages

- * No labeled required
- * Finds hidden patterns
- * Useful for preprocessing

Restricted Boltzmann Machine (RBM)

RBM is a probabilistic neural network consisting of:

- * Visible layer
- * Hidden layer

There are no connections within the same layer — only between visible and hidden units.

Applications

- * Recommendation systems
- * Collaborative filtering
- * Feature learning
- * Dimensionality reduction

Advantages

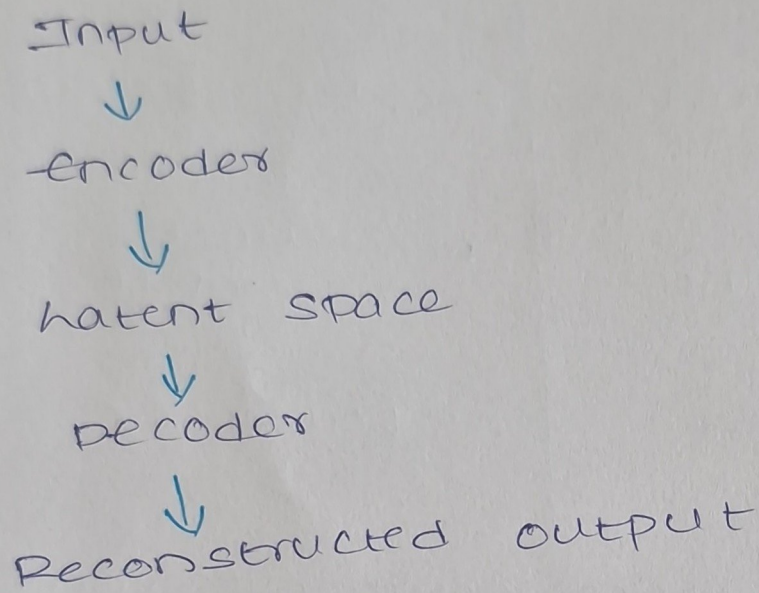
learns latent features

good for unsupervised poteroling

Autoencoders

An Autoencoder is a neural network trained to reconstruct its own input

Structure



Types

- Basic Autoencoder
- Sparse Autoencoder
- Denoising Autoencoder
- Variational Autoencoder

Applications

- ⇒ Image denoising
- ⇒ Image compression
- ⇒ Anomaly detection
- ⇒ Feature extraction
- ⇒ Data dimensionality reduction

RBM VS Autoencoders

Feature	RBM	Autoencoder
Learning Type	probabilistic	deterministic
Architecture	visible + hidden	encoder → decoder
Objective	learn probability distribution	Reconstruct input
Complexity	Higher	Lower
Common uses	Recommendation system, feature learning	compression, denoising, anomaly detection.